AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

Mathematics Paper II

Time: 2 hours 15 minutes Marks: 65

AKU May Examination 2017

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's signature**

2. RUBRIC. There are TEN questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use a black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. You may use a scientific calculator if you wish.

Q.1. (Total 6 Marks)

i. Find the real and imaginary parts of the complex number $Z = \left(\frac{3-2i}{4-3i}\right)^2$. (4 Marks)

ii. For $Z_1 = 13-12i$, prove that $|Z_1| = |\overline{Z}_1|$. (2 Marks)

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Q.2. (Total 8 Marks)

i. Find the matrix A in terms of a, if $\begin{bmatrix} a & a & a \\ a & a & a \\ a & a & a \end{bmatrix} = 2A - \begin{bmatrix} 0 & -a & -a \\ -a & 0 & -a \\ -a & -a & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}^{-1}$.

Also, identify the type of matrix A. (5 Marks)

ii. Without expansion, show that $\begin{vmatrix} 3 & 6 & 9 \\ 2 & 5 & 8 \\ -1 & 3 & 7 \end{vmatrix} = 0$. (3 Marks)

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Q.3. (Total 6 Marks)

If the 4th term of an arithmetic progression is 2 and the 9th term is $\frac{9}{2}$, then find the first term and the common difference. Also, find the 100th term of the progression.

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(ATTEMPT EITHER PART a OR PART b OF Q.4.)

Q.4. (Total 3 Marks)

a. Without using calculator, convert 0.481481481481481…. into an equivalent common fraction.

b. Find the sum of the series 1 + 4 + 9 +16 + 25 + 36 + …. + 2500.

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(ATTEMPT EITHER PART a OR PART b OF Q.5.)

Q.5. (Total 6 Marks)

a. Prove that $\binom{n}{3}+\binom{n}{2}=\binom{n+1}{3}$, where n is a positive integer.

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b. A non-transparent jar contains tickets numbered from 1 to 50. If two dice are rolled simultaneously and ticket is drawn from the jar, then answer the following questions.

i. Find the number of sample points in the sample space. (1 Mark)

ii. Are rolling a dice and drawing a ticket from the jar independent or dependent events? Justify your answer. (2 Marks)

iii. What is the probability of both dice showing the same number and ticket showing a number greater than 45? Also write all possible outcomes of the event. (2 Marks)

iv. Find the probability of the event that the sum on both dice is 9 and the ticket shows odd numbers. (1 Mark)

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Q.6. (Total 6 Marks)

i. Prove using mathematical induction that $5+9+13+17+.....+4n+1=n(2n+3)$ is true for all positive integral values of *n*. Support your working with necessary statements. (4 Marks)

ii. Find the third term in the expansion of $(2a-3b)^{10}$. (2 Marks)

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(ATTEMPT EITHER PART a OR PART b OF Q.7.)

Q.7. (Total 8 Marks)

a. Find the solution set of the equation $(x+1)(x+2)(x+3)(x+4)=3$.

b.

i. If the roots of equation $x^2+2x-15=0$ are α and β, then find the equation whose roots are $\alpha+\beta$ and $\frac{1}{\alpha}+\frac{1}{\beta}$.

ii. Apply synthetic division to find quotient and remainder when x^5-3x^3+x+7 is divided by $x-1$.

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(ATTEMPT EITHER PART a OR PART b OF Q.8.)

Q.8. (Total 7 Marks)

a.

i. Verify that $\frac{\left(\sin^2\theta+\cos^2\theta\right)\left(1+\cot^2\theta\right)}{\left(1+\tan^2\theta\right)}=\cot^2\theta.$ (4 Marks)

ii. The terminal ray of θ lies in the fourth quadrant and $\sin\theta=-\frac{4}{7}$. Without using the calculator, find the value of $\cot\theta$.

(3 Marks)

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b.

i. Apply formulae of $\sin(\alpha - \beta)$ and $\cos(\alpha - \beta)$ to deduce that $\tan(\alpha - \beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha \tan\beta}$.

(4 Marks)

ii. Verify that $\dfrac{\sin 3\alpha + \sin\alpha}{\cos 3\alpha - \cos\alpha} = -\cot\alpha$. (3 Marks)

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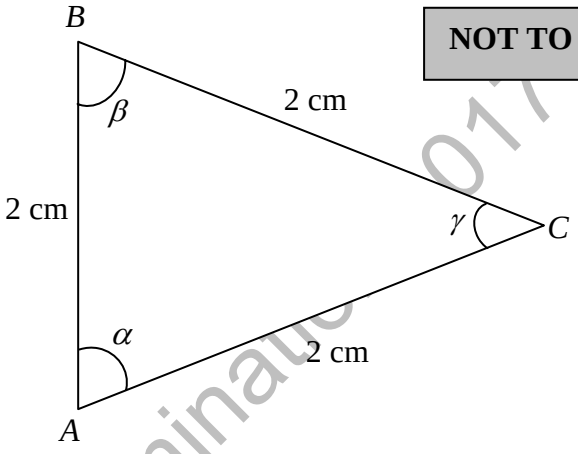
(ATTEMPT EITHER PART a OR PART b OF Q.9.)

Q.9. (Total 8 Marks)

a. In the given equilateral triangle *ABC*, the length of each side is 2 cm. For the given triangle *ABC*,

i. prove that $r_1 : R : r = 3 : 2 : 1$, where r_1 is the radius of the escribed circle opposite to the vertex A, R is the circum-radius and r is the in-radius. (6 Marks)

ii. find the value of $\sin\alpha$ where α is an angle associated with vertex *A*. (2 Marks)



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(ATTEMPT EITHER PART a OR PART b OF Q.9.)

b.

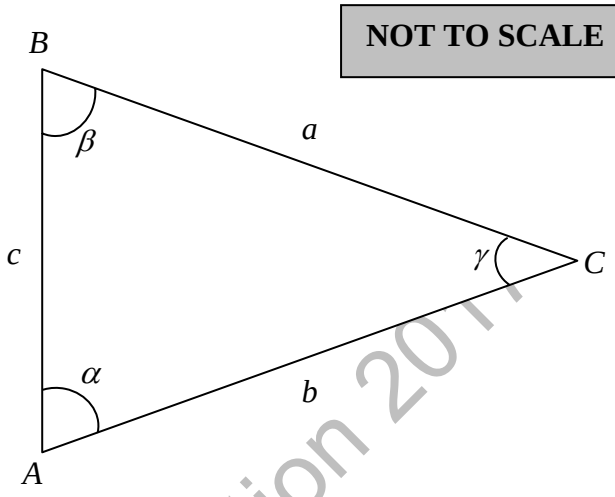
i. Prove the law of cosines i.e. $c^2 = a^2 + b^2 - 2ab\cos\gamma$ with the help of a suitable diagram. (6 Marks)

Space for diagram

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ii. In a triangle *ABC*, if the measure of $a = 6$ cm, $b = 7$ cm and $c = 3$ cm, then find the value of $\cos\frac{\alpha}{2}$. The semi perimeter *S* of the triangle *ABC* is equal to 8 cm. (2 Marks)

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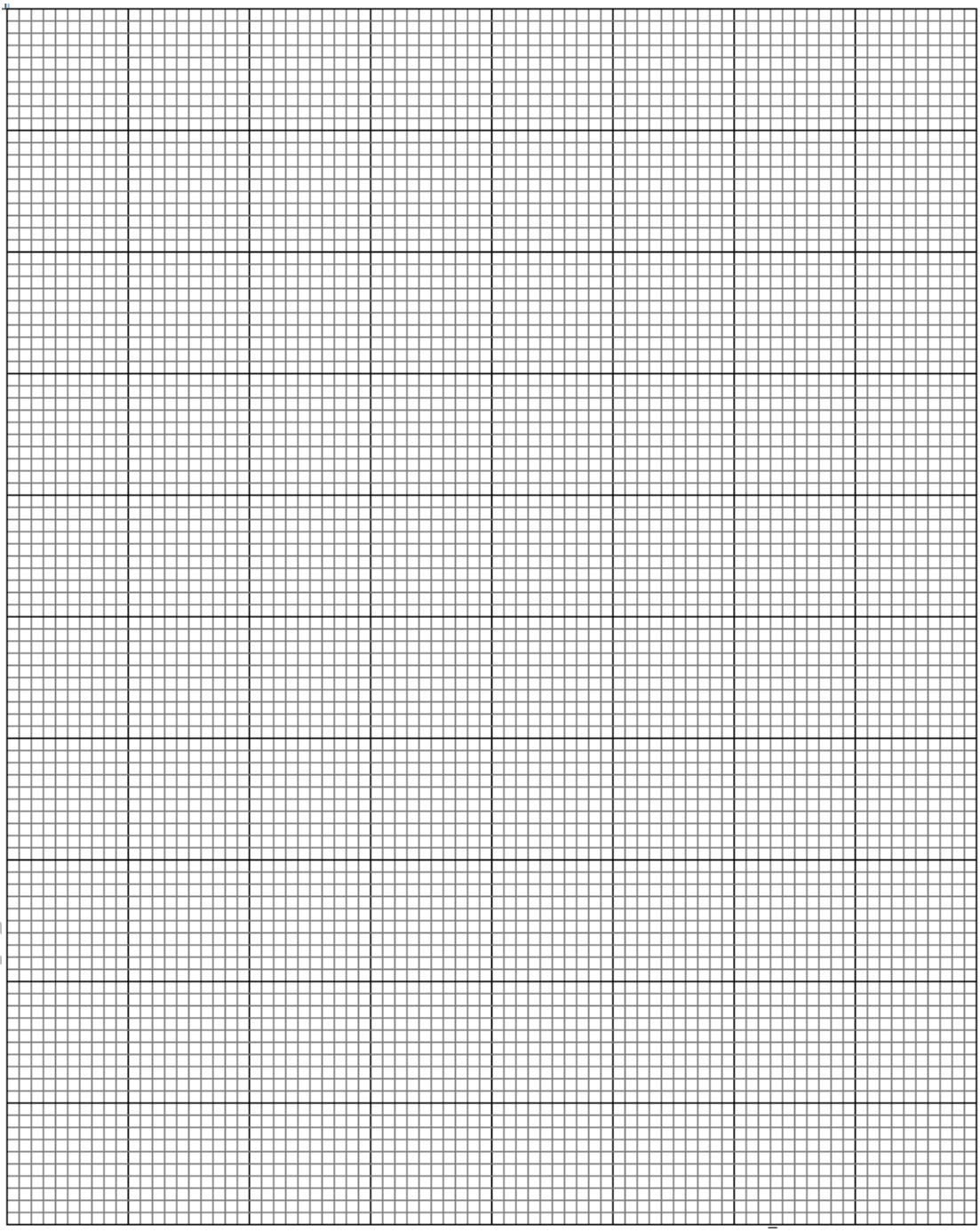
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(ATTEMPT EITHER PART a OR PART b OF Q.10.)

Q.10. (Total 7 Marks)

a. Fill the following table and draw the graph of $\sec\theta$ on the given graph. Write the range of the $\sec\theta$.

θ	$\sec\theta$
0	1
$\frac{\pi}{6}$	
$\frac{\pi}{3}$	
$\frac{\pi}{2}$	
$\frac{2\pi}{3}$	
$\frac{5\pi}{6}$	
π	
$\frac{7\pi}{6}$	
$\frac{4\pi}{3}$	
$\frac{3\pi}{2}$	
$\frac{5\pi}{3}$	
$\frac{11\pi}{6}$	
2π	



AK

b.

i. Solve $\cos x - \sqrt{3}\sin x = 0$. (3 Marks)

ii. Find the value(s) of x for the trigonometric equation $\cos 2x - \sin^2 x + 2 = 0$. (4 Marks)

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